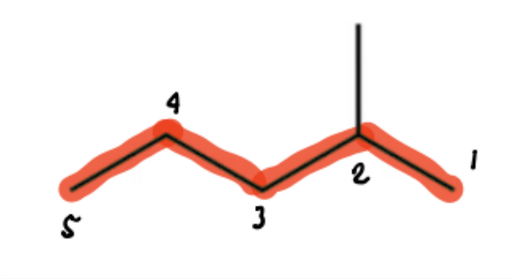
ACTIVITY 3 – NAME IT DRAW IT

ALRRIN R. ESCAPE

CHEMICAL ENGINEERING 2–A

CARD A01 - 2-methylpentane
Level: Foundation Molecular Formula: C_6H_{14}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 5 C, so the root is pentane. The methyl branch is on a top corner, so it is not part of the longest horizontal run. The path methyl–C2–C3–C4–C5 is also 5 C and leads to the same name.

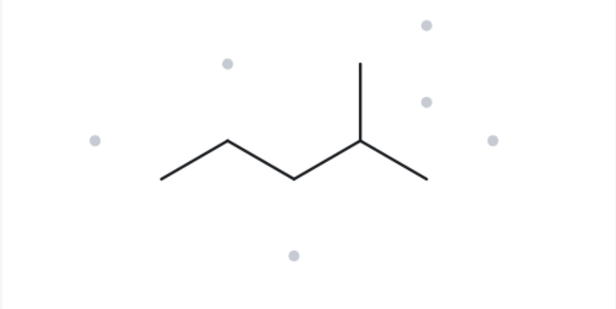
Numbering: left to right puts the methyl on C4, and right to left puts it on C2. The first point of difference is 2 < 4, so number from the right.

Substituents: methyl on C2, so 2-methyl.

3. *My Name*

2-methylpentane

4. *The Alkane Check*



2-methylpentane

Returned name: 2-methylpentane

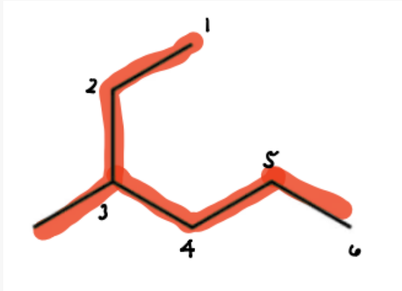
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A02 - 3-methylhexane
Level: Foundation Molecular Formula: C_7H_{16}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 6 C, so the root is hexane. It starts at the end of the upward arm, passes through the branch carbon, and continues along the lower zigzag. The path through the short down-left arm is only 5 C, so that arm is a substituent.

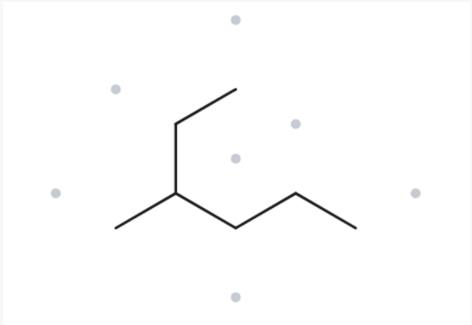
Numbering: numbering from the upper-left end puts the methyl on C3, and numbering from the right end puts it on C4. The first point of difference is 3 < 4, so number from the upper-left end.

Substituents: methyl on C3, so 3-methyl.

3. *My Name*

3-methylhexane

4. *The Alkane Check*



3-methylhexane

Returned name: 3-methylhexane

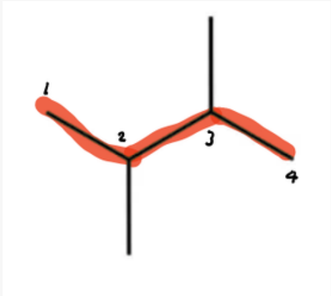
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A03 - 2,3-dimethylbutane
Level: Foundation Molecular Formula: C_6H_{14}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 4 C, so the root is butane. Several 4-C paths exist, but each one carries two methyl groups, so they all lead to the same name.

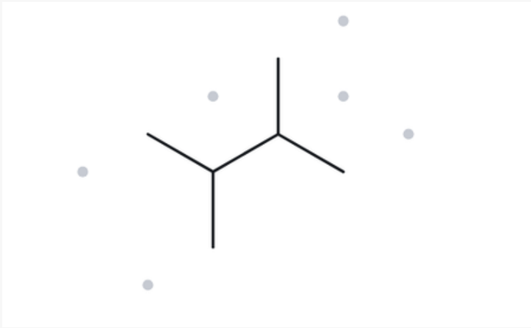
Numbering: left to right gives 2,3 and right to left also gives 2,3. The sets tie. There is only one kind of substituent, so no alphabetical tie-break is needed, and the name is the same either way.

Substituents: methyl on C2 and C3, so 2,3-dimethyl. Use the di- prefix with a comma between the numbers and a hyphen before the letters.

3. *My Name*

2,3-dimethylbutane

4. *The Alkane Check*



2,3-dimethylbutane

Returned name: 2,3-dimethylbutane

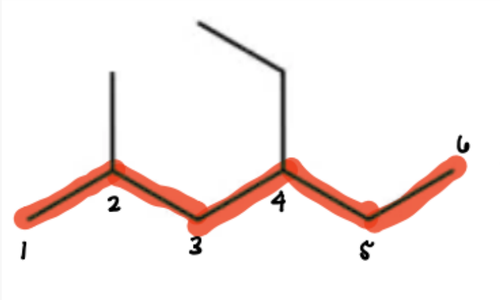
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A04 - 4-ethyl-2-methylhexane
Level: Developing Molecular Formula: C_9H_{20}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 6 C, so the root is hexane. Chains through the upward ethyl arm also reach 6 C, but they carry the same two substituents, so they lead to the same name. No chain reaches 7 C.

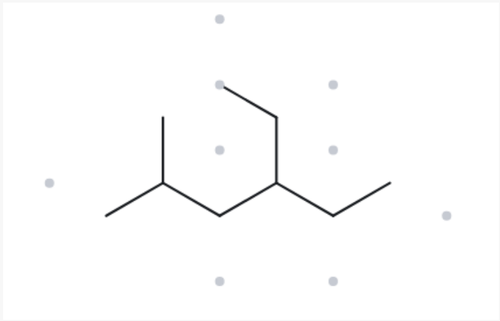
Numbering: left to right gives locants 2,4 (methyl on C2, ethyl on C4), and right to left gives 3,5. The first point of difference is 2 < 3, so number from the left.

Substituents: ethyl on C4, then methyl on C2. E comes before M, so ethyl is cited first even though its locant is higher.

3. *My Name*

4-ethyl-2-methylhexane

4. *The Alkane Check*



4-ethyl-2-methylhexane

Returned name: 4-ethyl-2-methylhexane

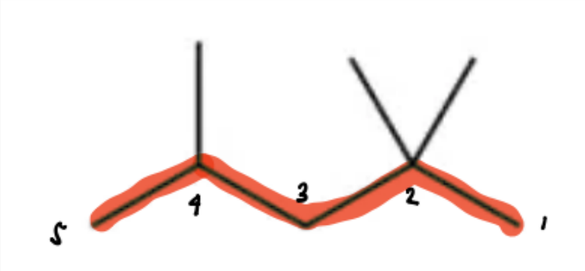
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A05 - 2,2,4-trimethylpentane
Level: Developing Molecular Formula: C_8H_{18}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 5 C, so the root is pentane. Several 5-C paths exist because of the two methyls on the same carbon, but each carries three methyl groups, so they lead to the same name. No chain reaches 6 C.

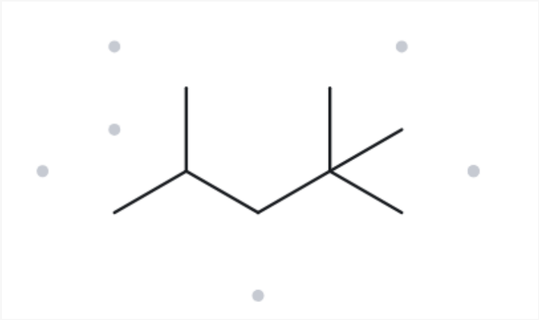
Numbering: left to right gives locants 2,4,4 and right to left gives 2,2,4. The first locants are both 2, so compare the next term: 2 < 4. The right-to-left set 2,2,4 wins at the second term, so number from the right.

Substituents: methyl on C2, C2 and C4, so 2,2,4-trimethyl. Use the tri- prefix, with one locant for each methyl (so 2 appears twice), commas between the numbers and a hyphen before the letters.

3. *My Name*

2,2,4-trimethylpentane

4. *The Alkane Check*



2,2,4-trimethylpentane

Returned name: 2,2,4-trimethylpentane

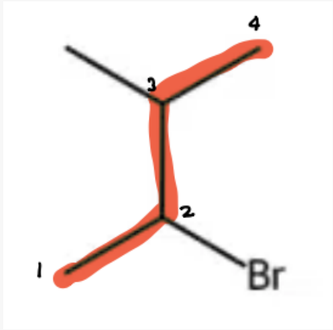
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A06 - 2-bromo-3-methylbutane
Level: Developing Molecular Formula: $C_5H_{11}Br$

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 4 C, so the root is butane. Several 4-C paths exist because of the two methyl arms on the branch carbon, but each carries one bromo and one methyl, so they lead to the same name. No chain reaches 5 C.

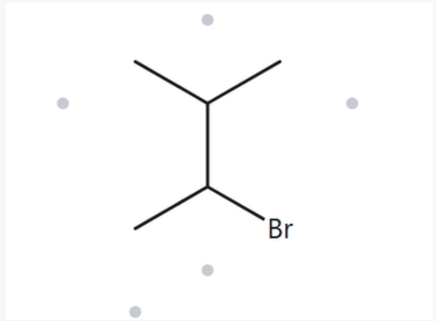
Numbering: left to right puts Br on C2 and methyl on C3, giving {2,3}. Right to left puts methyl on C2 and Br on C3, also {2,3}. The locant sets tie, so the substituent cited first alphabetically gets the lower number. Bromo (b) comes before methyl (m), so Br gets C2 and numbering runs from the left.

Substituents: bromo on C2, then methyl on C3. The halogen is a prefix here, named bromo.

3. *My Name*

2-bromo-3-methylbutane

4. *The Alkane Check*



2-bromo-3-methylbutane

Returned name: 2-bromo-3-methylbutane

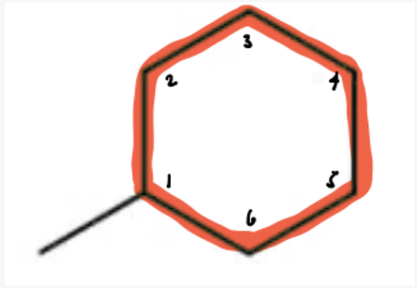
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A07- methylcyclohexane
Level: Developing Molecular Formula: C_7H_{14}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the ring has 6 C and the only chain is a 1-C methyl, so the ring is the parent (this activity's rule: a ring present is the parent). The root is cyclohexane.

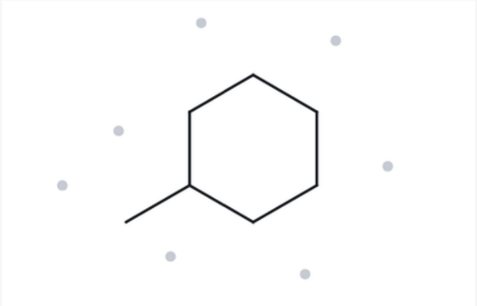
Numbering: with only one substituent, the ring carbon bearing it is C1 by default. Either direction round the ring gives the same single locant, so there is nothing to compare.

Substituents: methyl on C1. A ring with one substituent needs no locant, so the 1 is left out, as in the activity's quick reference.

3. *My Name*

methylcyclohexane

4. *The Alkane Check*



methylcyclohexane

Returned name:
methylcyclohexane

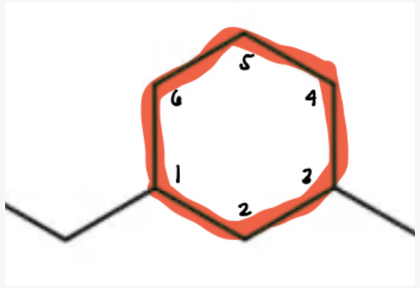
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A08- 1-ethyl-3-methylcyclohexane
Level: Challenge Molecular Formula: C_9H_{18}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the ring has 6 C and the longest chain is only the 2-C ethyl, so the ring is the parent. The root is cyclohexane.

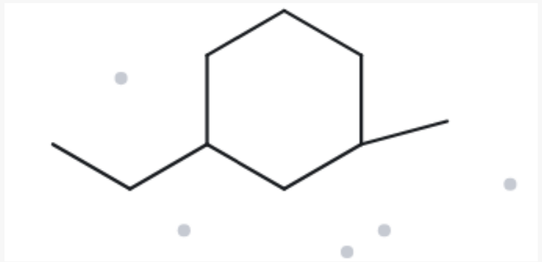
Numbering: on a ring, start at a substituent and go the shorter way round to the other one. The two substituents are separated by one ring carbon. Starting at the ethyl carbon gives {1,3}, and starting at the methyl carbon also gives {1,3}. The locant sets tie, so the substituent cited first alphabetically gets the lower number. Ethyl (e) comes before methyl (m), so ethyl is on C1 and methyl on C3.

Substituents: ethyl on C1, then methyl on C3.

3. *My Name*

1-ethyl-3-methylcyclohexane

4. *The Alkane Check*



1-ethyl-3-methylcyclohexane

Returned name: 1-ethyl-3-methylcyclohexane

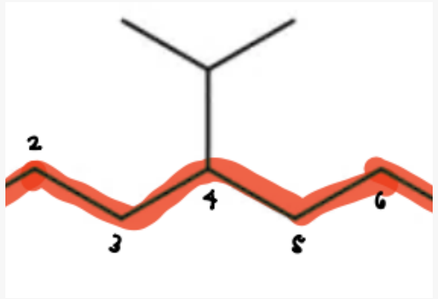
✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A09- 4-(1-methylethyl)heptane
Level: Challenge Molecular Formula: $C_{10}H_{22}$

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the longest chain is 7 C, so the root is heptane. It runs along the horizontal zigzag through the branch carbon, with three carbons on each side. The path that goes up through the branch is only 6 C (3 + the branch carbon + 2), so it isn't the parent even though it looks like a straight line in the drawing.

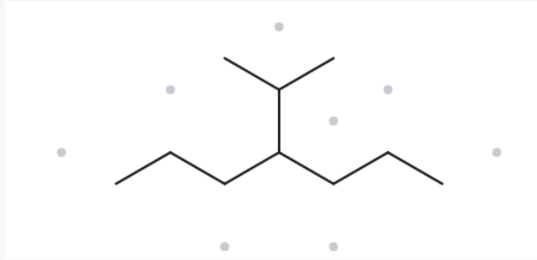
Numbering: left to right puts the branch on C4, and right to left also puts it on C4. The locants tie. With only one kind of substituent, no alphabetical tie-break is needed.

Substituents: the three-carbon branched group on C4 is isopropyl, written systematically as (1-methylethyl). Complex substituents go in parentheses, so the name is 4-(1-methylethyl).

3. *My Name*

4-(1-methylethyl)heptane

4. *The Alkane Check*



2-methyl-3-propylhexane

Returned name: 2-methyl-3-propylhexane

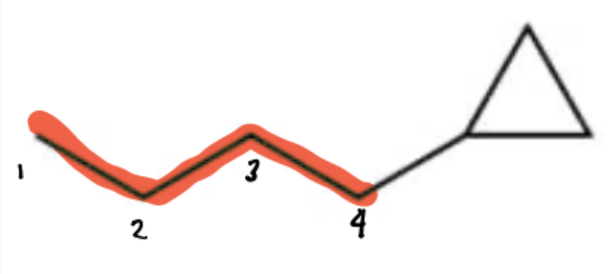
❌ Mismatch

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A10- butylcyclopropane
Level: Challenge Molecular Formula: C_7H_{14}

1. *Numbered Parent*



2. *Naming Trail*

Parent chain: the ring has 3 C and the chain has 4 C. In older textbooks the longer chain would win, but this activity uses the current IUPAC rule: when a ring is present, the ring is the parent. So the root is cyclopropane, and the 4-C chain is a substituent.

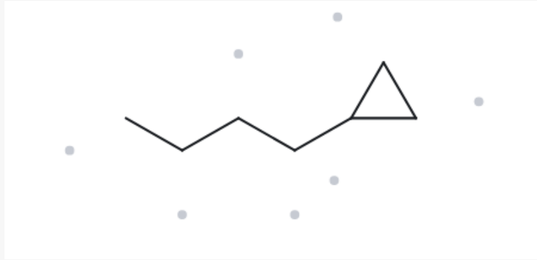
Numbering: there is only one substituent, so the ring carbon bearing it is C1. Either direction round the ring gives the same result, so there is nothing to compare.

Substituents: a four-carbon straight chain, which is butyl, on C1. A ring with one substituent needs no locant, so no “1-” is written.

3. *My Name*

butylcyclopropane

4. *The Alkane Check*



butylcyclopropane

Returned name:
butylcyclopropane

✓ Match

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

ACTIVITY 3 – NAME IT DRAW IT

PART B

CARD B01 - Given name: 2-methylpropane
Level: Foundation

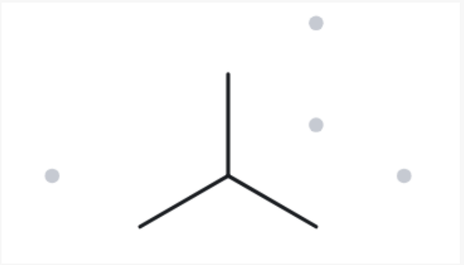
1. *Name Decoded*

Parent chain: propane, a 3-carbon chain.

Substituents: methyl on C2

Expected Formula: : 3 + 1 = 4 C. An open chain is C_nH_{2n+2}, so n = 4 gives C_4H_{10}.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_4H_{10}

Open chain: C_nH_{2n+2} = C_4H_{10}. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B02 - Given name: 3-ethylpentane
Level: Foundation

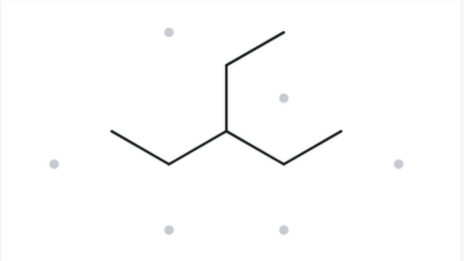
1. *Name Decoded*

Parent chain: pentane, a 5-carbon chain.

Substituents: ethyl on C3, the middle carbon.

Expected Formula: : 5 + 2 = 7 C. An open chain is C_nH_{2n+2}, so n = 7 gives C_7H_{16}.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_7H_{16}

Open chain: C_nH_{2n+2} = C_7H_{16}. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 3-ethylpentane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B03 - Given name: 2,2-dimethylbutane
Level: Foundation

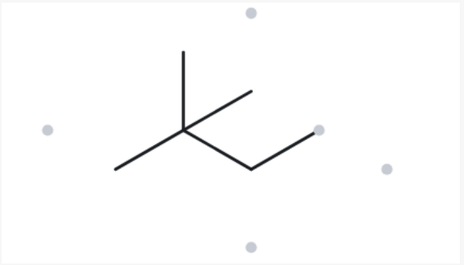
1. *Name Decoded*

Parent chain: butane, a 4-carbon chain.

Substituents: two methyl groups, both on C2 (the “2,2” and the di- prefix).

Expected Formula: : 4 + 2 = 6 C. An open chain is C_nH_{2n+2}, so n = 6 gives C_6H_{14}.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_6H_{14}

Open chain: C_nH_{2n+2} = C_6H_{14}. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 2,2-dimethylbutane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B04 - Given name: 3-ethyl-2-methylpentane
Level: Developing

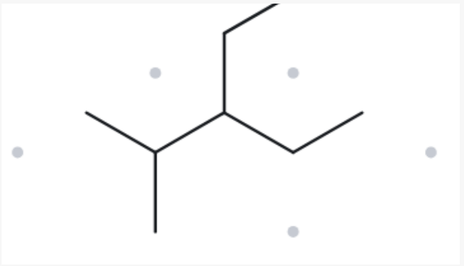
1. *Name Decoded*

Parent chain: pentane, a 5-carbon chain.

Substituents: methyl on C2 and ethyl on C3. They are on neighboring carbons.

Expected Formula: : 5 + 1 + 2 = 8 C. An open chain is C_nH_{2n+2}, so n = 8 gives C_8H_{18}.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_8H_{18}

Open chain: C_nH_{2n+2} = C_8H_{18}. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 3-ethyl-2methylpentane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B05 - Given name: 1-chloro-2-methylpropane
Level: Developing

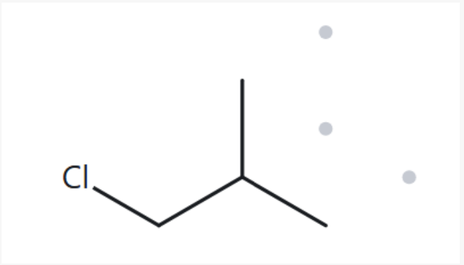
1. *Name Decoded*

Parent chain: propane, a 3-carbon chain.

Substituents: chloro on C1 and methyl on C2.

Expected Formula: : 3 + 1 = 4 C. An open chain is C_nH_{2n+2}, so n = 4 gives C_4H_{10}. Cl replaces one H, so the formula is C_4H_9Cl.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_4H_9Cl

Open chain: $C_nH_{2n+2} = C_4H_{10}$; minus one H for Cl = C_4H_9Cl

4. *Back Check*

Returned by The Alkane: 1-chloro-2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B06 - Given name: 1,2-dimethylcyclopentane
Level: Developing

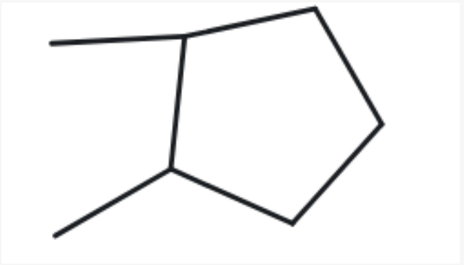
1. *Name Decoded*

Parent chain: cyclopentane, a 5-carbon ring.

Substituents: two methyl groups, on C1 and C2. These are adjacent ring carbons.

Expected Formula: : 5 + 2 = 7 C. One ring is C_nH_{2n}, so n = 7 gives C_7H_{14}.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_7H_{14}

Open chain: C_nH_{2n} = C_7H_{14}, no halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 1,2-dimethylcyclopentane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B07- Given name: 4-ethyl-3,3-dimethylheptane
Level: Developing

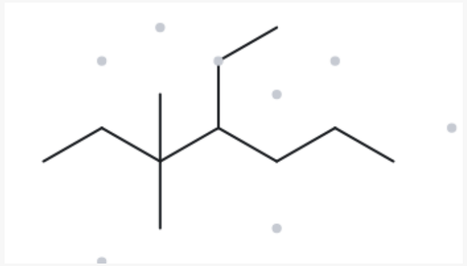
1. *Name Decoded*

Parent chain: heptane, a 7-carbon chain.

Substituents: ethyl on C4, and two methyl groups, both on C3 (the “3,3” and the di- prefix).

Expected Formula: : 7 + 2 + 2 = 11 C. An open chain is C_nH_{2n+2}, so n = 11 gives $C_{11}H_{24}$.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: $C_{11}H_{24}$

Open chain: C_nH_{2n+2} = $C_{11}H_{24}$. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B08 - Given name: tert-butylcyclohexane
Level: Developing

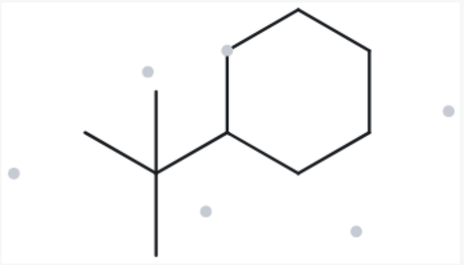
1. *Name Decoded*

Parent chain: cyclohexane, a 6-carbon ring.

Substituents: tert-butyl on the ring. It is a 4-carbon group with one central carbon bonded to three methyl groups, written $(CH_3)_3C–$. With a single substituent on a ring, no locant is needed.

Expected Formula: : 6 + 4 = 10 C. One ring is C_nH_{2n}, so n = 10 gives $C_{10}H_{20}$.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: $C_{10}H_{20}$

Open chain: C_nH_{2n+2} = $C_{10}H_{20}$. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane

The returned name is just the systematic IUPAC name of tert-butylcyclohexane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B09 - Given name: 3-ethyl-1,1-dimethylcyclohexane
Level: Developing

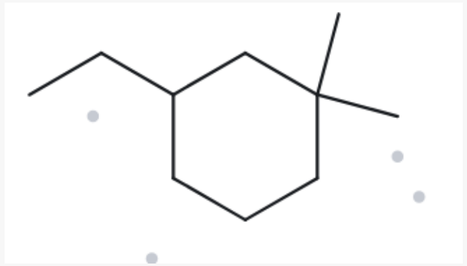
1. *Name Decoded*

Parent chain: cyclohexane, a 6-carbon ring.

Substituents: two methyl groups, both on C1 (the “1,1” and the di- prefix), and an ethyl on C3.

Expected Formula: : 6 + 2 + 2 = 10 C. One ring is C_nH_{2n}, so n = 10 gives $C_{10}H_{20}$.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: $C_{10}H_{20}$

Open chain: C_nH_{2n+2} = $C_{10}H_{20}$. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record

CARD B10 - Given name: 2-ethylbutane
Level: Developing

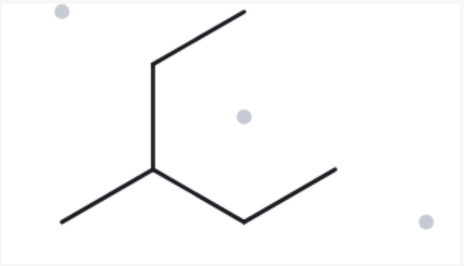
1. *Name Decoded*

Parent chain: pentane, a 5-carbon chain. The longest continuous chain runs through the ethyl group: $CH_3–CH_2–CH–CH_2–CH$

Substituents: methyl on C3.

Expected Formula: : 5 + 1 = 6 C. An open chain is C_nH_{2n+2}, so n = 6 gives C_6H_{14}.

2. *Bond-line Structure*



3. *Formula Check*

Counted from the drawing: C_6H_{14}

Open chain: C_nH_{2n+2} = $C_{10}H_{20}$. No halogens, so nothing is subtracted.

4. *Back Check*

Returned by The Alkane: 3-methylpentane

✗ Not the same as the given name

VERIFICATION STRIP

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check record